

Pilot Course on "Deep Generative Models: A Statistical Perspective"

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- 3) Course Schedule: Class meetings will be held between 10:00 AM and 12:00 PM on the following dates in Room 108-319(E) (Lunch will be provided):

04/24 (F), 05/08 (F), 05/29 (F), 06/12 (F), 06/19 (F), 07/03 (F), 07/24 (F), 07/31 (F)
- 4) Office Hours: 1:00 PM - 2:00 PM every Wednesday in Room 108-301-11 until the end of the pilot course (July 31st). Discussions on any topics related to generative models, not just the pilot course material, are welcome, so please feel free to stop by.
- 5) Recording: The pilot lectures will be recorded (only the instructor and the lecture slides will be captured).
- 6) Group Presentation:
 - Group Presentation (to be held on 07/31): Students will select a tutorial related to generative models from top-tier AI conferences such as ICML and NeurIPS. Using these materials, a flipped-learning style group presentation will be conducted.

7) Detailed Schedule:

Week	Contents	Remark
01	<u>04/24 (F): Pilot Lecture 1:</u> Course Introduction + Fundamentals of Probability Theory + Random Variables	* Midterm Exam Period
02	-	
03	<u>05/08 (F): Pilot Lecture 2:</u> Statistical Estimation and Hypothesis Testing	* 05/05 (T): Children's Day
04	-	
05	-	
06	<u>05/29 (F): Pilot Lecture 3:</u> Basics of Probabilistic Modelling	
07	-	
08	<u>06/12 (F): Pilot Lecture 4:</u> Autoregressive Model + Energy-based Models	
09	<u>06/19 (F): Pilot Lecture 5:</u> Variational Autoencoders	* Final Exam Period
10	-	
11	<u>07/03 (F): Pilot Lecture 6:</u> Generative Adversarial Networks	
12	-	
13	-	
14	<u>07/24 (F): Pilot Lecture 7:</u> Other Statistical Distances-based Approaches	
15	<u>07/31 (F): Pilot Lecture 8:</u> Group Presentations	<u>Group Presentation</u>